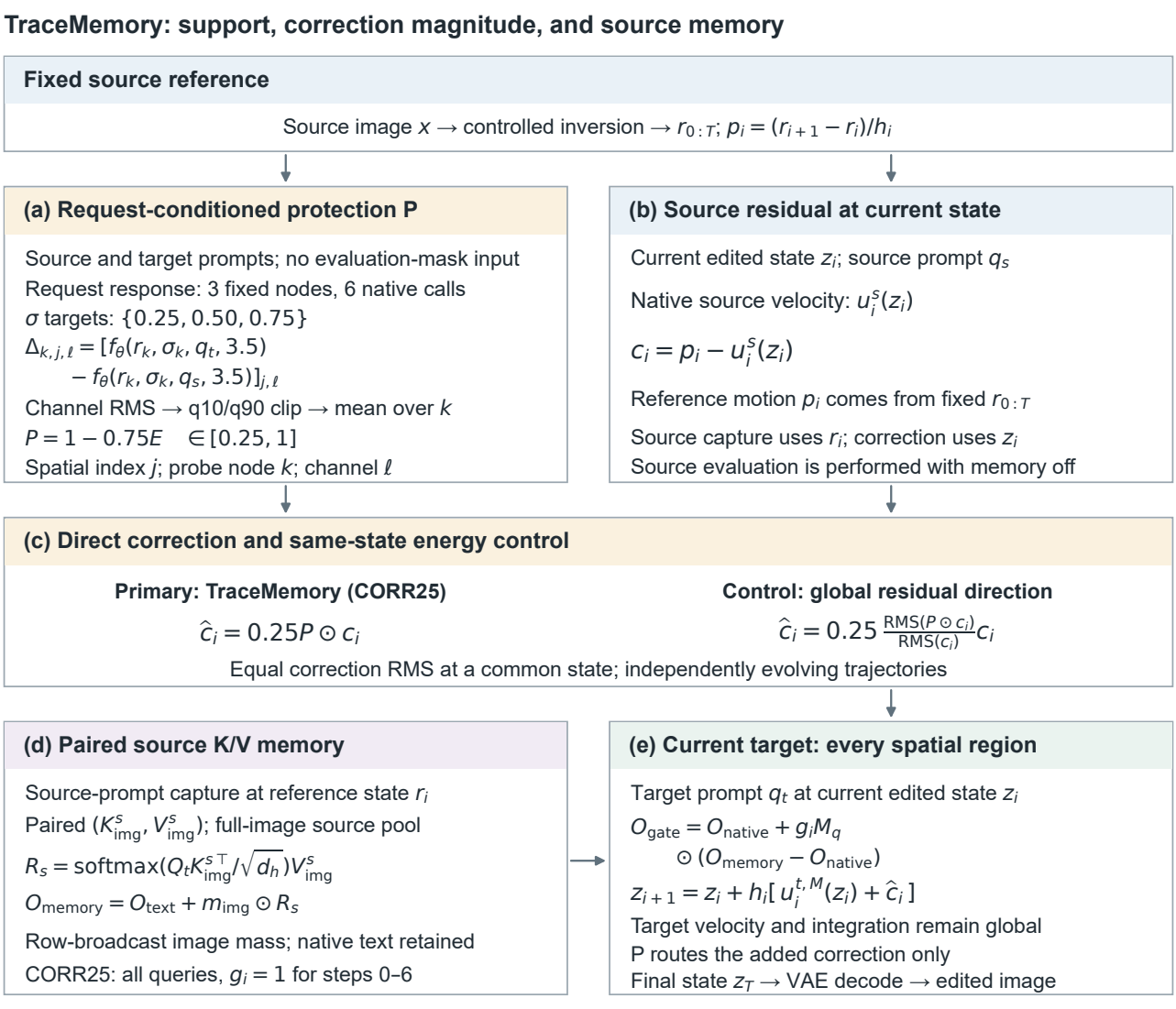

TraceMemory: support, correction magnitude, and source memory
Fixed source reference
Source image x → controlled inversion → $r_{0:T}$; $p_i = (r_{i+1} - r_i)/h_i$
(a) Request-conditioned protection P
Source and target prompts; no evaluation-mask input
Request response: 3 fixed nodes, 6 native calls
σ targets: $\{0.25, 0.50, 0.75\}$
$\Delta_{k,j,\ell} = [f_\theta(r_k, \sigma_k, q_t, 3.5)$
$- f_\theta(r_k, \sigma_k, q_s, 3.5)]_{j,\ell}$
Channel RMS → q10/q90 clip → mean over k
$P = 1 - 0.75E \quad \in [0.25, 1]$
Spatial index j; probe node k; channel ℓ
(b) Source residual at current state
Current edited state z_i; source prompt q_s
Native source velocity: $u_i^s(z_i)$
$c_i = p_i - u_i^s(z_i)$
Reference motion p_i comes from fixed $r_{0:T}$
Source capture uses r_i; correction uses z_i
Source evaluation is performed with memory off
(c) Direct correction and same-state energy control
Primary: TraceMemory (CORR25)
$\hat{c}_i = 0.25 P \odot c_i$
Control: global residual direction
$\hat{c}_i = 0.25 \frac{\mathrm{RMS}(P \odot c_i)}{\mathrm{RMS}(c_i)} c_i$
Equal correction RMS at a common state; independently evolving trajectories
(d) Paired source K/V memory
Source-prompt capture at reference state r_i
Paired $(K^s_{\mathrm{img}}, V^s_{\mathrm{img}})$; full-image source pool
$R_s = \mathrm{softmax}(Q_t K^{s\top}_{\mathrm{img}} / \sqrt{d_h}) V^s_{\mathrm{img}}$
$O_{\mathrm{memory}} = O_{\mathrm{text}} + m_{\mathrm{img}} \odot R_s$
Row-broadcast image mass; native text retained
CORR25: all queries, $g_i = 1$ for steps 0–6
(e) Current target: every spatial region
Target prompt q_t at current edited state z_i
$O_{\mathrm{gate}} = O_{\mathrm{native}} + g_i M_q$
$\odot (O_{\mathrm{memory}} - O_{\mathrm{native}})$
$z_{i+1} = z_i + h_i[\, u_i^{t,M}(z_i) + \hat{c}_i \,]$
Target velocity and integration remain global
P routes the added correction only
Final state z_T → VAE decode → edited image